Advanced Statistics

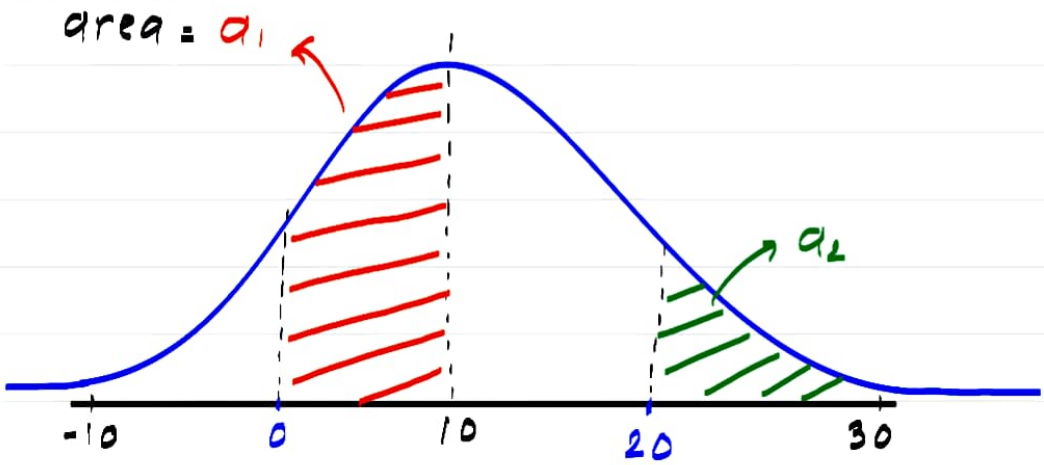
23-01-26

Normal distribution

- its a continuous distribution
- also known Gaussian distribution
- it describes distribution of continuous data where most of the data is clustered around central values forming a symmetrical bell shaped curve.

- it gives overall characteristics of data such as
  - central tendency
  - dispersion
  - extreme values / outliers
  - skewness

- Using normal dist. we can easily identify proportion of data lying within given range.



total area under any KDE = 1

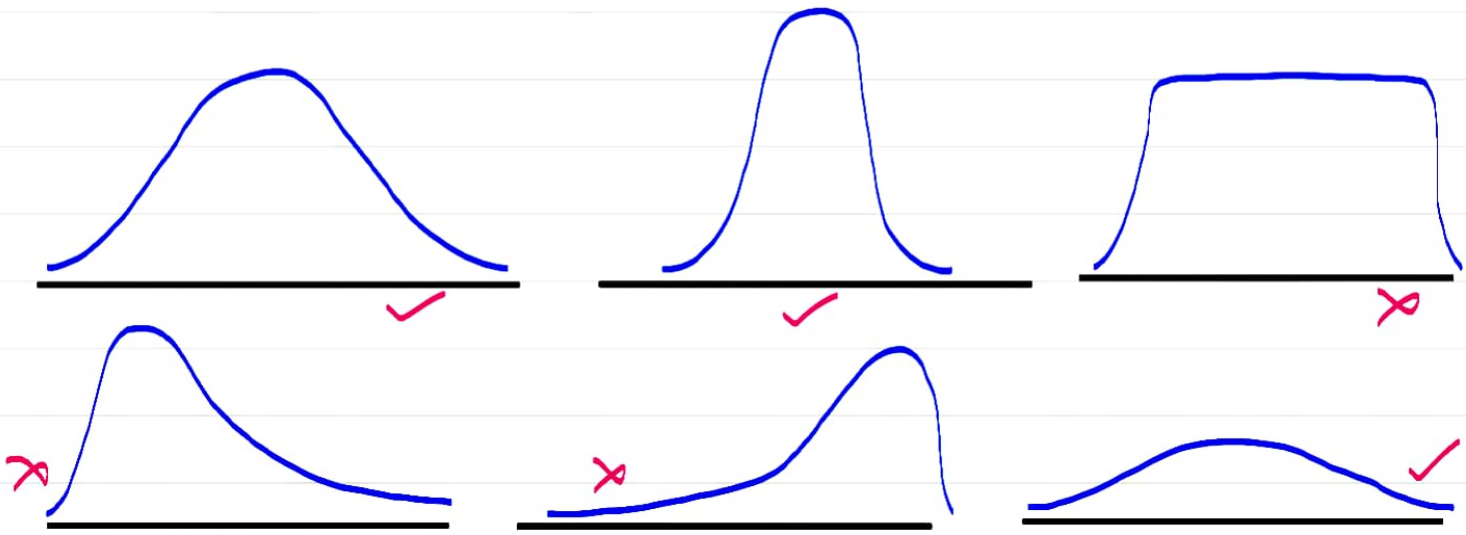
whats proportion of data lying betn 0 - 10 ?
→ a_1

whats proportion of data lying betn 20-30 ?
→ a_2

based on the area under the curve

proportion of data betn 0-10 > proportion of data betn 20-30

A lot of natural phenomena and processes follow <u>normal distribution</u>



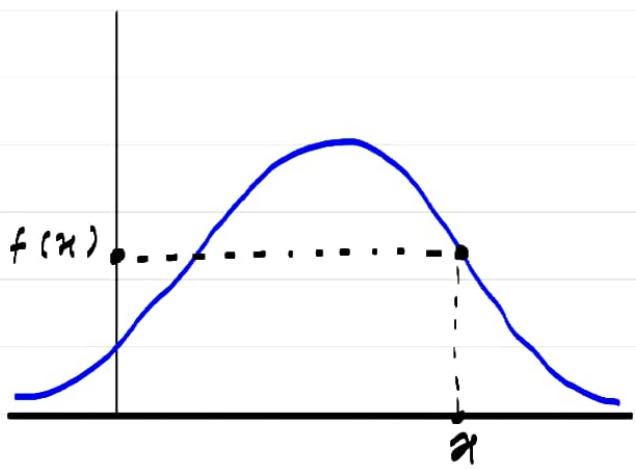
- When a KDE plot is a nearlly symmetrical bell shaped curve then its a normal distribution

- Normal dist. has two main parameters - mean and std dev

- Inferential stats is heavily based on properties of normal dist.
  eg. central limit theorem

PDF of normal distribution

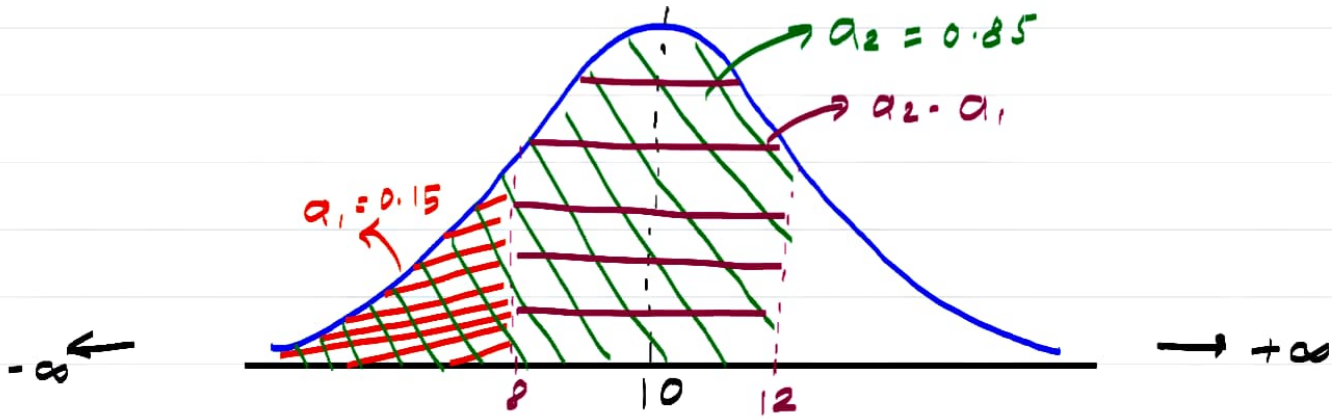
$$f(x) = N(\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\left[\frac{(x-\mu)^2}{2\sigma^2}\right]}$$



- rather than density, we are more interested in area under the curve for normal distribution

- CDF, cummulative density function gives the area under the curve.

Q. - A dataset follows normal dist with mean = 10 and std dev = 3.
find the proportion of data lying betn 8 to 12



cdf(x) = area under the curve from $-\infty$ to x

cdf ($x = 8$) = a_1 = 0.15

cdf ($x = 12$) = a_2 = 0.85

area under curve from 8 to 12 = $a_2 - a_1$
= cdf(12) - cdf(8)

- proportion of data lying betn 11 to 14

= cdf (14) - cdf(11)

Exponential distribution

- its a continuous distribution

- used to give probability of occurence of next event, where the events are occuring at given rate.

- its used to find
  - time betn arrival of customers
  - lifespan of a product
  - time untill machine fails

eg.

1) In a shop on an averge 10 customers visit every 1 hr. The time betn customers arrival follows exponential dist. Whats probability that the next customer visits within 20 mins?

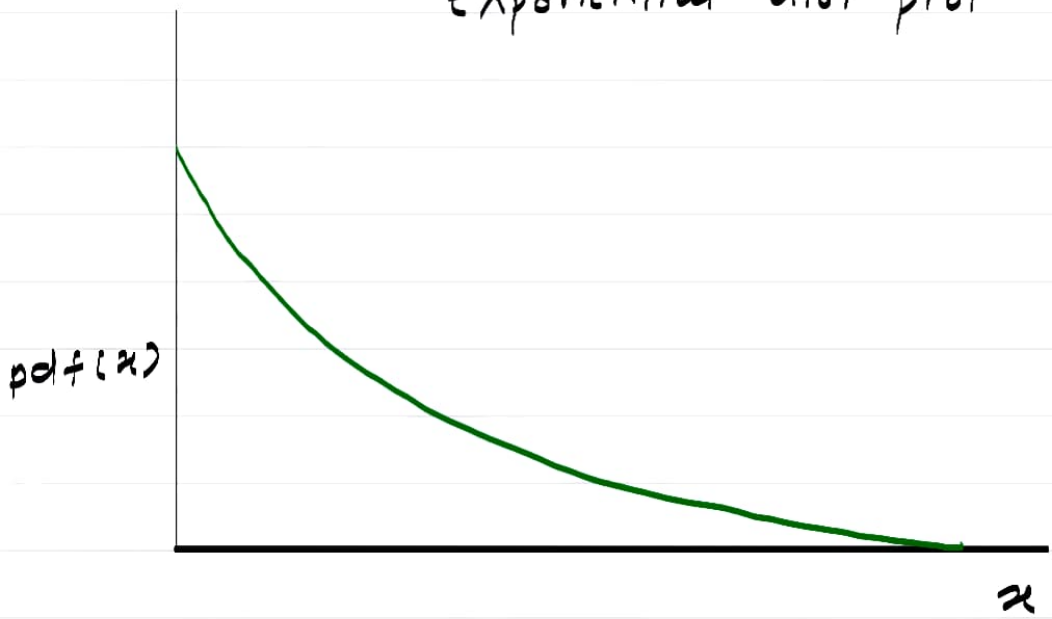
2) if a machine fails once every 4 hrs on an average time interval betn failures follows exponential dist. Whats prob. that next failure occurs before 2hrs?

PDF

$$f(x, \lambda) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

$$f(x, \lambda) = 0 \quad \text{for } x < 0$$

exponential dist plot



exponential dist has one parameter- λ

λ is average number of events per unit time

CDF

$$cdf(x, \lambda) = 1 - e^{-\lambda x} \quad \text{for } x \geq 0$$
$$cdf(x, \lambda) = 0 \quad \text{for } x < 0$$

Ans for questions on top

1)

$\lambda = 10$ customers every hr

$\lambda = \frac{10}{60}$ customers every minute $= \frac{1}{6}$

prob. that next customer visits before 20 mins
$x \leq 20$

we need prob for $x \leq 20$ $\Rightarrow$ area under curve where $x \leq 20$

area under curve for $x \leq 20$ $= cdf(x=20) = 1 - e^{-\lambda x} = 1 - e^{-(\frac{1}{6} \times 20)}$

2)

$\lambda = 1$ failure every 4 hrs

$\lambda = \frac{1}{4}$ failure every 1 hr

prob that next failure occurs before 2hrs

Prob of $x \leq 2$ = cdf $(x=2)$ = area under the curve

$$\text{cdf}(x=2) = 1 - e^{-\lambda x}$$

$$= 1 - e^{-(\frac{1}{4} \times 2)}$$

$$= 1 - e^{-1/2} = 1 - \frac{1}{\sqrt{e}} = 1 - \frac{1}{\sqrt{2.71}}$$